

TABLE OF CONTENT

NORMATIVE REFERENCES

DEFINITIONS

DESIGNATIONS AND ABBREVIATIONS

INTRODUCTION

- Relevance of the Research
- Scientific Novelty
- Research Goal
- Research Objectives
- Object and Subject of Research
- Research Hypothesis
- Methodological Basis
- Provisions Submitted for Defense
- Theoretical Significance
- Practical Significance
- Approbation of Research Results
- Publications

1 PROBLEM DOMAIN ANALYSIS AND CURRENT STATE OF AUTOMATED DIABETIC RETINOPATHY DIAGNOSIS

- 1.1 Medical and Epidemiological Context of Diabetic Retinopathy
 - 1.1.1 Pathophysiology and Clinical Grading Systems
 - 1.1.2 Screening Requirements in Resource-Limited Healthcare Settings
- 1.2 Fundus Image Acquisition and Quality Variability
 - 1.2.1 Sources of Image Degradation in Clinical Practice
 - 1.2.2 Impact of Image Quality on Diagnostic Model Performance
- 1.3 Deep Learning Approaches to Retinal Image Classification
 - 1.3.1 Convolutional Neural Network Architectures for Medical Imaging
 - 1.3.2 Transfer Learning Strategies in Ophthalmic Diagnostics
- 1.4 Critical Analysis of Existing Automated DR Screening Systems
- 1.5 Formulation of the Research Problem and Justification of Research Direction
- Conclusions to Chapter 1

2 THEORETICAL FOUNDATIONS OF IMAGE PREPROCESSING AND DEEP LEARNING FOR FUNDUS IMAGE ANALYSIS

- 2.1 Mathematical Foundations of Image Enhancement Techniques
 - 2.1.1 Histogram Equalization and Adaptive Contrast Enhancement
 - 2.1.2 Formalization of CLAHE with Controllable Threshold Parameters
 - 2.1.3 Spatial Filtering and Noise Reduction Methods
- 2.2 Theoretical Framework of Convolutional Neural Networks
 - 2.2.1 Convolution, Pooling, and Feature Extraction Operations

- 2.2.2 Loss Functions and Optimization for Imbalanced Medical Datasets
- 2.2.3 Regularization Techniques: Dropout, Batch Normalization, and Data Augmentation

2.3 Transfer Learning Theory and Domain Adaptation

- 2.3.1 Feature Transferability Across Visual Domains
- 2.3.2 Frozen-Layer versus Progressive Fine-Tuning Strategies

2.4 Mathematical Modeling of Laser-Tissue Interaction in Retinal Therapy

2.4.1 Coupled Thermal-Optical Model of Fundus Tissue Response

2.4.2 Implications for Diagnostic Image Feature Interpretation

Conclusions to Chapter 2

3 METHODOLOGY OF INTEGRATED PREPROCESSING-CNN PIPELINE DESIGN

3.1 Formalization of the Unified Preprocessing Pipeline

- 3.1.1 Pipeline Stage Specification: Resizing, Normalization, Enhancement, Augmentation
- 3.1.2 Modified CLAHE Algorithm with Simplified Threshold Control
- 3.1.3 Augmentation Strategy for Class Imbalance Mitigation

3.2 Design of Baseline and Enhanced CNN Architectures

- 3.2.1 Shallow Baseline CNN for Preprocessing Quality Validation
- 3.2.2 Enhanced Multi-Block CNN with Regularization Layers

3.3 Transfer Learning Methodology Using EfficientNetB0 and ResNet50

- 3.3.1 Architecture Adaptation for Five-Class DR Classification
- 3.3.2 Two-Stage Fine-Tuning Protocol Design
- 3.3.3 Weighted Loss Function Formulation for Ordinal Class Structure

3.4 Evaluation Framework and Performance Metrics

- 3.4.1 Multi-Metric Assessment: Accuracy, F1-Score, ROC-AUC, Cohen's Kappa
- 3.4.2 Cross-Validation and Statistical Reliability Protocols

Conclusions to Chapter 3

4 EXPERIMENTAL RESEARCH: PREPROCESSING IMPACT ON CNN DIAGNOSTIC PERFORMANCE

4.1 Datasets and Experimental Configuration

- 4.1.1 APTOS 2019, STARE, and Supplementary Clinical Image Corpora
- 4.1.2 Class Distribution Analysis and Data Partitioning Strategy
- 4.1.3 Hardware Constraints and Computational Resource Limitations

4.2 Experiment 1: Baseline CNN without Preprocessing versus Enhanced CNN

- 4.2.1 Training Dynamics and Convergence Analysis
- 4.2.2 Quantitative Comparison of Diagnostic Metrics

4.3 Experiment 2: Modified CLAHE Threshold Optimization on Small Datasets

- 4.3.1 Threshold Parameter Sensitivity Analysis
- 4.3.2 Impact on Feature Preservation in Microaneurysms and Small Vessels

4.4 Experiment 3: Transfer Learning Strategy Comparison

- 4.4.1 EfficientNetB0: Frozen versus Progressive Fine-Tuning

- 4.4.2 ResNet50: Feature Extraction versus End-to-End Fine-Tuning
- 4.4.3 Per-Class Performance Analysis under Severe Class Imbalance

Conclusions to Chapter 4

5 RELIABILITY VALIDATION AND COMPARATIVE ANALYSIS

5.1 Cross-Database Generalization Testing

- 5.1.1 Model Transferability Across Heterogeneous Image Sources
- 5.1.2 Stability Assessment under Varying Image Quality Conditions

5.2 Statistical Validation of Preprocessing Dominance Hypothesis

- 5.2.1 Ablation Study: Preprocessing Components versus Architectural Complexity
- 5.2.2 Quantitative Evidence for Image Quality as Primary Performance Driver

5.3 Comparative Analysis with Existing DR Diagnostic Systems

- 5.3.1 Benchmarking Against Published Results: IDx-DR, EyeNuk, DeepMind
- 5.3.2 Performance-Complexity Trade-Off Analysis

5.4 Limitations and Boundary Conditions of the Proposed Approach

Conclusions to Chapter 5

6 ARCHITECTURE OF AN AUTOMATED DR SCREENING SYSTEM FOR RESOURCE-LIMITED ENVIRONMENTS

6.1 System Requirements and Design Principles

- 6.1.1 Functional and Non-Functional Requirements Specification
- 6.1.2 Modular Architecture with PACS and EHR Integration

6.2 AI Processing Module Design

- 6.2.1 Preprocessing Engine with Configurable Pipeline Parameters
- 6.2.2 Inference Module with Model Selection Logic

6.3 Clinical Workflow Integration

- 6.3.1 Telemedicine and Portable Device Support for Rural Deployment
 - // Deployment in distributed telemedicine systems
 - // Integration with national eHealth platforms
 - // Real-time remote DR screening in low-resource regions
- 6.3.2 Physician-in-the-Loop Decision Support Interface

6.4 Data Security and Regulatory Compliance Framework

- 6.4.1 GDPR/HIPAA-Aligned Data Management Protocols
- 6.4.2 Applicability to Kazakhstan Healthcare Infrastructure

Conclusions to Chapter 6

CONCLUSION

REFERENCES

APPENDICES

- Appendix A — Source Code of the Preprocessing Pipeline
- Appendix B — Supplementary Experimental Results and Confusion Matrices
- Appendix C — System Architecture UML Diagrams

Appendix D — Certificates of Implementation and Approbation Acts